A PROJECT REPORT

on

Resume Optimizer with Llamaindex

A Report submitted for the Special Project Program

B.Tech 4th Year

In

Computer Science & Engineering

To

THE ICFAI UNIVERSITY TRIPURA

by

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Course Title: Special Project-II

Course Code:- CSE452

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Declaration of Student

We hereby declare that the project entitled **"Resume Optimizer with Llamaindex"** submitted for the 'Special Project' Program of B.Tech (CSE) 4th year degree is my original work and the project has not formed the basis for the award of any other degree, diploma, fellowship or any other similar titles.

SIGNATURE: -

Student Name:-

Joy Majumder (22IUT0010111)

<u>CERTIFICATE</u>

The ICFAI University, Tripura

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING CERTIFICATE

This is to certify that the project titled **"Resume Optimizer with Llamaindex" is** the bonafide work carried out by Joy Majumder student of B.Tech(CSE) of Department of Computer Science and Engineering, The ICFAI University, Tripura during the Summer Term-2025, in partial fulfilment of the requirements for the award of the degree of B.tech CSE and that the project has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

Signature of the Guide

Place:

Date:

ABSTRACT

In today's competitive job market, a well-optimized resume is essential for capturing the attention of recruiters and Applicant Tracking Systems (ATS). The **Resume Optimizer with LlamaIndex** is an AI-powered application designed to intelligently analyze and enhance resumes based on specific job descriptions. Built using **Streamlit** for an interactive user interface and **Ollama with LlamaIndex** for natural language processing, this system provides personalized, data-driven recommendations to improve resume effectiveness.

The tool allows users to upload their PDF resumes and input job titles and descriptions to receive **targeted optimization suggestions** across multiple dimensions including **ATS keyword optimization**, **experience section enhancement**, **skills hierarchy creation**, **professional summary crafting**, **education optimization**, and **career gap framing**. Using **LlamaIndex**, the system constructs contextual knowledge representations of both the resume and the job post, enabling precise comparison and recommendation generation.

By leveraging modern language models and structured data analysis, the Resume Optimizer assists job seekers in creating resumes that are not only more readable and impactful but also more likely to pass automated screening systems. The result is a powerful, end-to-end resume enhancement solution that bridges the gap between human readability and machine compatibility empowering users to present their qualifications with greater clarity, relevance, and confidence.

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APPROVAL SHEET

The project report entitled **"Resume Optimizer with Llamaindex"** by Joy Majumder (22IUT0010111) the student of B.Tech 4th Year Semester-II , has approved for the Special Project Program in Computer Science and Engineering at ICFAI University, Tripura:

Examiners:

1. Dr. Saptarshi Chakraborty ______________________________

2. Prof. Durgabati Podder ______________________________

3. Prof. Sukanya Salia ______________________________

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5. Dr. Debapriya Banik ______________________________

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INTRODUCTION

1.1) PROBLEM DEFINITION:

Crafting a resume that effectively aligns with a job description is a major challenge for job seekers. Most candidates struggle to tailor their resumes to specific job postings and fail to pass through Applicant Tracking Systems (ATS). Manual optimization is time-consuming and prone to human bias. Therefore, there is a need for an AI-powered system that can intelligently analyze resumes and provide targeted suggestions to enhance their relevance and structure.

1.2) PURPOSE :

The purpose of this project is to design and develop an intelligent resume optimization tool that uses AI to help job seekers improve their resumes based on job-specific requirements. The system aims to make resumes both **ATS-friendly** and **recruiter-focused**.

1.3) AIM AND OBJECTIVES:

Aim:

To develop an AI-based Resume Optimizer that enhances the structure, relevance, and readability of resumes based on job postings.

Objectives:

1. To extract and process resume text from PDF files.
2. To analyze job titles and descriptions using LlamaIndex.
3. To provide specific optimization modes such as keyword enrichment, experience enhancement, and skills organization.
4. To offer real-time feedback and AI-generated suggestions for resume improvement.

CHAPTER 2

LITERATURE SURVEY

2.1) EXISTING SYSTEM:

In today's digital world, numerous resume enhancement tools and online builders exist to assist job seekers in improving their resumes. However, most of these systems rely on **rule-based matching** or **simple keyword extraction**, which often fails to capture the semantic context between job descriptions and resume content.

While they can highlight missing terms or generic improvements, such tools typically lack **AI-driven contextual understanding**, resulting in limited personalization and accuracy.

1.Structure of Existing Systems:

Most traditional resume analyzers include:

a. A resume parser that extracts text from PDF or Word documents.
b. A keyword matcher that compares extracted text against job description terms.
c. A scoring algorithm that assigns a match percentage based on keyword overlap.

These systems generally perform the following tasks:

a. Extract resume text and metadata.
b. Compare extracted data with job requirements or skill lists.
c. Generate a basic score or keyword report for user reference.

2.Vulnerabilities in Existing Systems:

Although these tools are functional, they fail to analyze contextual relationships between job roles, experiences, and skills.
They depend heavily on keyword frequency rather than meaning, resulting in inaccurate feedback for candidates with varied terminology or non-standard experience descriptions.

3.Common Weaknesses:

a. No Contextual Understanding:
Tools often ignore semantic relationships—e.g., treating "software developer" and "programmer" as unrelated.

b. Limited Customization:
Most systems provide generic advice, not tailored recommendations based on a specific job description.